

SECTION: CRISIS AND INNOVATION

CRISIS, INNOVATION, AND CHANGE MANAGEMENT IN LESS-DEVELOPED LOCAL BUSINESS ECOSYSTEMS: THE CASE OF EASTERN MACEDONIA AND THRACE

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ABSTRACT: This study explores the co-evolutionary triangle of crisis, innovation, and change management in less-developed local business ecosystems. It takes as a case study the region of Eastern Macedonia and Thrace - one of the least developed regional socioeconomic business ecosystems in Greece and the EU - and tries to find out the correlation between the different aspects of this “global crisis-innovation-change management” triangle from the perspective of a sample of micro and small enterprises operating in this ecosystem. To this end, the authors conducted a qualitative, introductory, field research, finding that there is a significant gap between the predominant scientific view of regional underdevelopment and the perception of the people of everyday business practice. This article supports that this deficient perception is one of the determining factors of the delayed development for this regional entrepreneurial ecosystem.

JEL CLASSIFICATIONS: O10, O30, R11, R58

KEYWORDS: Regional underdevelopment, global structural crisis, weak regional business ecosystems, less-developed regions, innovation, change management, globalization’s restructuring

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1. Positioning the problem: crisis, innovation, and change management in less-developed regional socio-economic systems

The current crisis and restructuring of globalization have not emerged “in the vacuum.” It has developed and matured structurally upon the former globalization development regime. This regime seems to have exhausted the limits of its previous socio-economic development potential. The current structural adjustment of globalization is, in essence, a transformation of the last regime accumulation of world capitalism, where new production and consumption conditions and arrangements are progressively replacing the old ones (Bresser-Pereira, 2010; Laudicina & Peterson, 2016). However, the current situation seems

to include many shadowy points that the conventional economic theory struggles to interpret sufficiently. This insufficiency is also the main reason why relatively new research trends in economics seem to gain an ever-stronger interest (Boyer & Saillard, 2002; Dopfer & Potts, 2009; Durand, 2000; Georgescu-Roegen, 1971; Heinzel, 2013; Passet, 1996).

In introductory terms, this article argues that this “new global age” that struggles to emerge nowadays, amid various detents and opposing views, different “sides” and perspectives, predestinates a new phase of globalization which has not yet been fully formulated (Bhattacharya, Khanna, Schweizer & Bijapurkar, 2017; Braudel, 2014). This new phase of globalization is a further evolution phase in the development of global capitalism, where the “newborn” tries to come forth and survive upon the “old” regime. This new state of affairs attempts to open a period of overall relative balanced growth globally (Ocampo, 2011; Pieterse, 2012; Rodrik, 2011).

In this context, all actors and actions at all levels-local, national, international, regional, and global scale- seem to evolutionary co-create and co-transform with their response the formation of this new global economy (Maroufkhani, Wagner, & Wan Ismail, 2018; Muller, 2013; Peneder, 2017; Zahra & Nambisan, 2011). In this process, nothing seems to be fully specified and evolutionarily unambiguous: The partial socio-economic components of this new phase of globalization, within a dialectic process of “theses-syntheses-antitheses” (Morabito, Sack, & Bhate, 2018; Norrie, 2009), are trying to impose their choices, to survive, grow and dominate evolutionarily.

This struggle for dominance in this new era of globalization and the consequent change in the rules of the “capitalist game” seems to tend to change drastically, at the same time, the architecture of the political economy discipline and the applied economic policy at all levels (Boyer, 2015; Schmidt, 2018; Van Den Bergh & Kallis, 2013). Indeed, it seems to become progressively more evident to policymakers that the central dialectic node in this overall relocation process is the “living” capitalist firm and the innovative action which articulates, diffuses and reproduces (Anderson, Potočník, & Zhou, 2014; Damanpour & Aravind, 2012).

Under these circumstances, the less-developed socio-economic systems perceive and react to this global restructuring with a variety of perceptions, ways, and mechanisms. In particular, they seem to view innovation according to their historically-defined way (Pitelis & Runde, 2017) and the context of their specific developmental trajectory (Andreoni & Scazzieri, 2014). Therefore, there are significantly different, and predestinated within the system, ways to manage the generated change (By, Burnes, & Oswick, 2012).

With these clarifications in mind, the question raised by this article includes the following specific aspects:

- How does the literature understand the triangle of crisis, innovation, and change management for less-developed regional socioeconomic systems?
- How are these three dimensions explained analytically?
- How do the business-people perceive in the context of the regional entrepreneurial system of Eastern Macedonia and Thrace the spheres of crisis, innovation, and change management nowadays?

The following structure presents how this study is going to approach these specific dimensions and queries methodologically. Initially, it presents the literature and analyzes the scientific contributions related to the aspect of regional underdevelopment and poor business ecosystems in conditions of crisis. Then, it approaches the analytical triangle of crisis, innovation deficiency, and incomplete change management. Taking as an empirical field test the less-developed regional business ecosystem of Eastern Macedonia and Thrace, it presents consequently a field survey carried out in a sample of small and micro enterprises in this particular region. It discusses the findings related to the surveyed firms' perception of the aspects of the crisis, innovation, and change management. Finally, it reaches into conclusions and discusses their analytical limitations and possible future advancements of this research.

2. Literature review: regional underdevelopment and poor business ecosystems in crisis conditions

There is recent scientific literature review that provides a variety of definitions for the case of less-developed regions (Askenazy & Martin, 2015; Szerb, Lafuente, Horváth, & Páger, 2018). With a particular focus on the technological aspect of regional development and underdevelopment and exploring the change of innovation process models as an opportunity to break out of path dependency for less developed regions, Pylak (2015) argues that less-developed regions fall within the low-tech model called "primary-sector-based manufacturing regions" and face difficulties when phasing into higher-tech models.

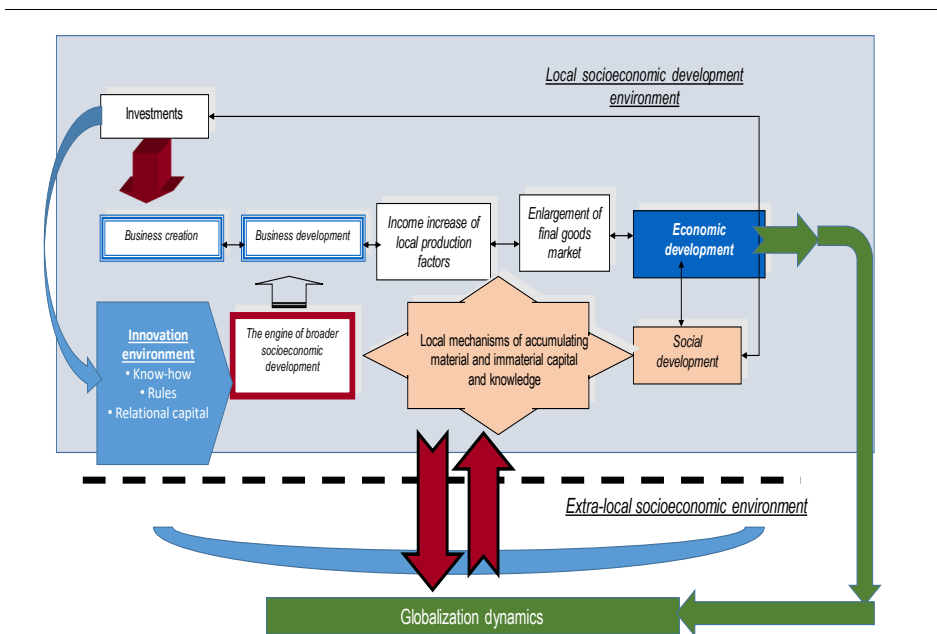
Gaffikin & Morrissey (2001) distinguish between declining and prosperous regions assuming that by definition, the regional economies are smaller than their nation-states and, as a result, they are more open and more dependent on external trade. The success requires an ability to attract external customers, while their labor markets are also more open-the declining regions tend to lose knowledge-intensive and highly skilled workers, while prosperous regions gain them.

Smętkowski (2018), who explores the role of exogenous and endogenous factors in the growth of regions in Central and Eastern Europe, invokes the theory of cumulative causation-which was formulated in the mid-20th century by Myrdal (1957) and then formalized by Kaldor (1970). In this context, spatial disequilibrium should increase because of the interplay of free market forces. Therefore, fertile regions become even more vibrant and poor ones more miserable because of a vicious circle of causes. The reason for this is that highly developed regions are attractive to workers since they offer better salaries, especially to those with relevant qualifications. Due to the inflow of a new workforce, the local labor market develops, thus encouraging new enterprises to invest and attracting inward capital and this, in turn, boosts production and, thanks to the benefits of agglomeration, increases productivity, thereby strengthening the region's competitive advantage and encouraging the influx of more workers. On the contrary, adverse external effects-associated, for example, with the depletion of the region's resources-can hinder the progression of this model; they can lead to "deglomeration," that is, the spatial diffusion of development.

From their side, Trippl, Asheim, & Mörner (2016), by exploring how to identify regions with less-developed research and innovation systems, conclude that a Regional Innovation System (RIS) can be seen as less-developed if it is ill-equipped to generate innovations along existing industrial and technological paths. However, this system might also be less-developed in the sense that it cannot support the renewal of the regional economy over time.

Overall, a significant amount of the contemporary literature on underdevelopment seems to converge in highlighting the primary internal dimensions as the leading causes of the underlying regional underdevelopment. The shortcomings and deficiencies in terms of retaining highly skilled human resources, insufficient assimilation of sophisticated business models and poor performance in terms of growth capacity of strong competitive dynamics seem to be the main interpretive dimensions for regional underdevelopment. Therefore, an innovation environment approach that favors the process of enhancing innovation capacities on a local scale systematically seems the most proper way to enhance the overall adaptability and reproduction of competitiveness of the socio-economic formations and, as a result, to develop within globalization (Figure 1).

FIGURE 1. INNOVATION ENVIRONMENT AND LOCAL DEVELOPMENT.



Source: Based on Vlado, Denizos, Chatzinikolaou, & Digkas (2019)

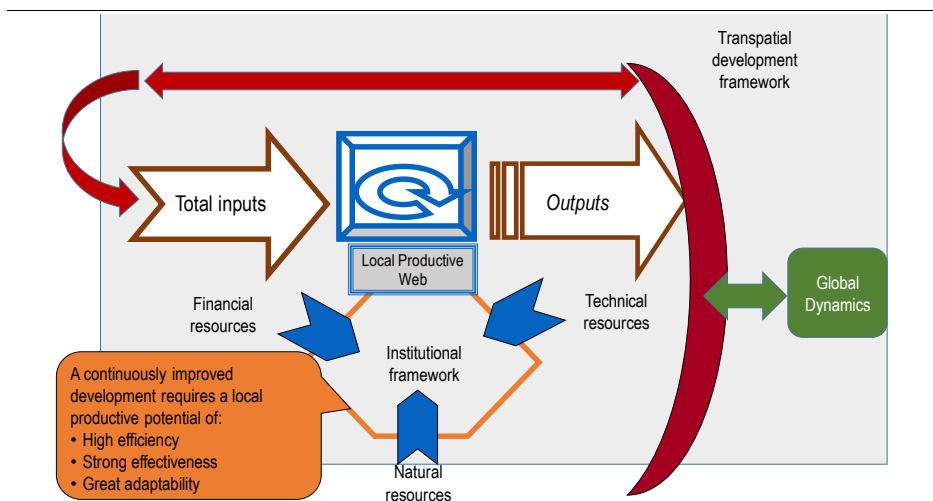
In this perspective, precisely, there are contemporary research studies that seem to focus and enrich the theory of local development in terms of local innovation capacity (Dauth & Suedekum, 2016).

Moreover, the subject of weak but also durable business ecosystems that includes the dynamics of clusters emerges. This subject also examines the aspects of regional underdevelopment and development. According to Harrison, Scheela, Lai, & Vivekarajah (2018), who explore the institutional voids that constrain the behavior of economic agents and by focusing on the emerging “business angel” market in Malaysia, a lack of a matching platform to make the connection between investors and entrepreneurs results in a poorly developed entrepreneurial ecosystem.

Spigel & Harrison (2018) argue that the existing research on entrepreneurial ecosystems has been mostly typological and atheoretical and has not yet explored how they influence the entrepreneurship process. The authors conclude that less-developed ecosystems may see an outflow of resources as entrepreneurs realize they must leave the region to grow their firm because of a lack of available investment capital, demands from investors that they relocate, or the need to move to broader labor markets to tap the talent they need.

One of the major disseminators of the concept of business ecosystems, Moore (2015), argues that the ecosystem as a whole must draw deeply from advances in science and engineering. The author notes in particular that the lesson of these cases for the connected community is that it is a long-term, patient, and complicated process: but one that is essential to the advances of the community. Without science and engineering input, the whole community stagnates.

FIGURE 2. A DEVELOPMENTAL APPROACH THAT FOCUSES ON THE EFFECTIVE MANAGEMENT OF LOCAL MATERIAL AND IMMATERIAL RESOURCES.



Source: Based on Vlados, Deniozos, Chatzinikolaou & Digkas (2019).

Consequently, lansiti & Levien (2004), in their widely-cited book “The Keystone Advantage: What the New Dynamics of Business Ecosystems Mean for Strategy, Innovation, and Sustainability”, which suggests that just as “keystone species” play central roles in their

biological ecosystems large companies such as Walmart and Microsoft deploy “keystone strategies” and shape and regulate their business ecosystems. Based on this, the authors suggest that networked innovation strategies are not without risks and, in particular, the biggest challenge for a niche player is to find itself in an ecosystem characterized by weak platforms or in an environment in which its platforms become in some way endangered.

The contemporary literature about less-developed business ecosystems appears to converge mainly to the lack of resource of innovative entrepreneurship as a determinant factor that leads to a weak evolutionary business ecosystem. In particular, the lack of capital investment, the need to move to broader labor markets to tap the talent the investors need, as well as the shortage in science and engineering input, largely determine the widening of the growth lag. Furthermore, it seems that to meet local development needs is critical to strengthen and exploit the local potential of each area effectively, in the current context of globalization dynamics (Figure 2).

2.1. Less-developed regions and crisis

Nowadays, the research on the phenomenon of crisis within less-developed regions and business ecosystems is also critical. The relevant approaches and interpretations show a significant variety. Eriksson & Hane-Weijman (2017) analyze how regional economies respond to crises and suggest that although the geography of resistance to crises and the ability of adaptability in the aftermath vary, cohesive (i.e., with many skill-related industries) and diverse (i.e., with a high degree of unrelated variety) regions are more resilient over time. They also find that resistance to future shocks (e.g., the 2008 recession) is highly dependent on the resistance to previous crises. By extension, this suggests that the long-term evolution of regional economies also influences their future resilience.

Salvati (2016), who refers to the Greek case, suggests that the recession between 2008-2012 has widened the gap between the capital Athens and the rest of Greece, while it has mitigated the disparities in per capita income between rich and poor regions. The author concludes that this evidence seems to contrast with literature indicating how dynamic regions came to be less exposed to recession than economically disadvantaged regions. The author finds that the 2008-09 crisis stimulates a rethinking of the geography of socio-economic disparities in highly divided countries, pointing out how expansion and recession differently affect the spatial distribution of wealth.

Omstedt (2016), who examines England’s North-South divide, by following a perspective the analyzes the partial phenomena of the crisis, claims that although the crisis emerged in a spatially concentrated financial sector, its impacts will be worse in places less associated with that sector. Therefore, the author argues that the crisis will hit mostly the more disadvantaged regions and localities of the country.

Cuadrado-Roura, Martin, & Rodríguez-Pose (2016), who examine the urban and regional consequences of the 2007-2008 economic crisis in Europe, suggest that while contrasts in wealth and productivity remain significant, the crisis has unearthed a considerable diversity in the capacity of regions and cities to adapt to shocks and respond to challenges.

Therefore, they conclude that this has also added new facets to how scholars and policymakers approach the problems in Europe's regions.

Therefore, it looks like the crisis of a local system, and its inadequacy in terms of competitiveness is the two sides of the same coin. In this context, the competitiveness of each spatially established socioeconomic formation appears to require a systematic increase in attractiveness nowadays (Malecki, 2017). At the same time, the attractiveness of a spatially defined system cannot be evolutionarily viable without a "mechanism" capable of building/rebuilding the competitive advantages of the firm hosted in the socio-economic environment.

This way, a local socio-economic system to be competitive must be able to attract the sustainable and innovative business interest of global reach. In this context, to attract this business interest, any socio-economic system must demonstrate how it can adequately accommodate the competitive ambitions of its components from a global perspective. Besides, the innovative business interest generates, maintains, produces, and reproduces, the competitive dynamics of the local system (Avdeychik, 2014; Covi, 2016).

2.2. The problem of innovation in less-developed regional business ecosystems

In the literature on the underdevelopment of less-developed regional business ecosystems, the overall problem of innovation appears to occupy a central position. According to Rodríguez-Pose & Wilkie (2019), who explore innovation capacity of less-developed regions and, in particular, what drives patenting in the lagging regions of Europe and North America, several factors constrain the innovative capacity of less-developed areas. Most prominent among them are socioeconomic and institutional deficiencies and geographic isolation. The authors argue that these contextual deficiencies relate to the weakness of the "local economic fabrics," the insufficient stocks of human and physical capital, and the absence of the formal and informal institutions that would typically function as the backbone of "innovation prone" environments.

According to Brown & Mason (2017), there is a critical review and conceptualization of entrepreneurial ecosystems, suggesting that the majority of observers view ecosystems primarily as a spatial concept to explain why certain places have high levels of entrepreneurial activity. This way, rather than having innovation at its core, entrepreneurship is the fundamental driver behind the concept. They conclude that the misconceptions about the nature and functioning of entrepreneurial ecosystems create the potential for misconceived policy interventions, signified by the continued emphasis on start-ups and the lack of genuinely systemic policy instruments. They are noticing that, given their pervasive heterogeneity, there is unlikely to be a "one-size fits all" policy prognosis for developing different types of ecosystems. Finally, scholars need to dissect further, conceptualize, theorize, and empirically examine this complex phenomenon much more closely to move the understanding of this problem forward.

In their turn, Liguori, Bendickson, Solomon, & McDowell (2019) attempt to develop a multi-dimensional measure for assessing entrepreneurial ecosystems. They argue that researchers and theorists have put considerable effort into defining and examining entrepreneurial ecosystems and business clusters development in certain regions. They claim in particular that favorable entrepreneurship ecosystems drive business and innovation. However, a commonly accepted measure of entrepreneurial ecosystem favorableness is absent. In this context, they search to build a psychometrically robust measure to capture entrepreneurs' perceptions of the surrounding ecosystem.

Finally, Walsh & Winsor (2019) provide insight into a region impeded from embracing the benefits of innovation-driven entrepreneurship in fostering economic development. The authors show that socio-cultural factors may be inhibiting the region from having a functional entrepreneurial ecosystem that can support innovation. They identify specific aspects of culture and social capital weaknesses and search for the potential causes of these impediments. Moreover, they conclude that the fundamental nature of culture may be affecting other elements of the entrepreneurial ecosystem from maturing.

Overall, in the context of the current restructuring phase of globalized capitalism, all actors try to produce new ways of resolving past or new problems in their competitive struggle: that is, trying to innovate. This element becomes crucial for the development of less-developed business ecosystems around the world. In this context, the issue of articulating a development strategy at each level (micro, meso, or macro; Dopfer, 2011; Vladoš & Katimertzopoulos, 2018) in the current restructuring of globalization (Pieterse, 2011) ceases to concern the mechanisms of national reach exclusively. Every socioeconomic formation-in this case, a less-developed socioeconomic business ecosystem-at every level (local, regional, or national) must now design and implement its unique, coherent developmental strategy. Moreover, the "innovative capitalism" seems to require an ever-increasing reproduction, diffusion, and assimilation of knowledge dynamics for all the socio-economic "actors" involved and hosted in all the partial socio-economic systems (Argote, 2012; Asheim, Grillitsch, & Trippel, 2017). To a great extent, therefore, the dimensions of innovation and knowledge are taking a central position within new theoretical approaches of spatial development (Boschma & Frenken, 2009). In this direction, the ability to innovate and the change management mechanisms are becoming a durable engine of spatial development inside globalization (Serrat, 2017).

In this sense, the co-evolving "crisis/development-innovation-change management" triangle can provide a new theoretical platform to understand the developmental phenomenon comprehensively-at all levels of space and, in particular, at the less-developed regional business ecosystems.

2.3. Change management and policies to foster entrepreneurship in less-developed regions

Change management and policies to foster entrepreneurship as scientific disciplines did not particularly concern scholars of local development until recently. However, there is contemporary literature that discusses this dimension increasingly-and especially in terms of

strengthening local entrepreneurship subsystems. North & Smallbone (2006), who explore how policy-makers can achieve the development of entrepreneurship and enterprise in Europe's peripheral rural areas, argue that a more strategic and coordinated approach towards building the entrepreneurial capacity of peripheral rural areas is necessary, based on a clearer vision of the role that enterprise can play in future rural development.

Huggins & Williams (2011), who explore whether the regional policy in less competitive regions that accounts for issues relating to entrepreneurship and enterprise development is a tool for improving regional competitiveness, find that entrepreneurship policy at the regional level is multidimensional-policies are ranging between the economically or the socially driven. They conclude that, although there is a significant policy activity in these areas across less competitive regions, enterprise policymaking remains relatively undifferentiated across the regions. Therefore, several developments in regional policy take place, especially towards shifting from policies relating to the facilitation of clusters to those focusing on developing regional innovation ecosystems. In extension, they find that regional policymakers are under pressure to measure short-term outputs at the expense of long-term nurturing.

Williams & Vorley (2014) outline the links between economic resilience and entrepreneurship. After studying the case of Sheffield City Region of England, suggest that enterprise, particularly as it relates to entrepreneurs and small firms, is crucial to economic resilience. In particular, the ability of entrepreneurs and small businesses to be adaptable and flexible is paramount to their capacity to absorb and respond to external shocks that in turn has a positive impact on the resilience of their locality. Besides, they conclude eloquently that, in more lagging regions, such as traditional industrial areas, weaker adaptation can occur due to embedded path dependency and low levels of existing entrepreneurship.

Horlings & Padt (2013) explore the aspect of leadership for sustainable regional development in rural areas. They examine the “underestimated” human factor in regional development and argue that sustainable development is far more than achieving environmental goals. It is rooted in inner motivations about a better life for the generations to come without undermining ecological resilience. As they argue, despite the significant amount of literature on change management in organizations, these approaches are too limited to explain how to bring about sustainable regional development for two reasons. Firstly, leaders are required to lead in multi-actor regional networks beyond the boundaries of organizations. Secondly, leadership for sustainable development is dealing with many visions and trying to align people around a joint sustainability vision. They find in particular that it is not about realizing ego-driven goals, but about adapting to the higher goal of required societal change, i.e., sustainable development.

Overall, concerning change management in public policy, and according to Kuipers et al. (2014), researchers could improve the theory-building on change management in public organizations with more and stronger empirical research that builds on a clear understanding of the practice. They would also need to pay more attention to the outcomes and successes of change in public organizations and to support practitioners in their search for lessons on what makes a change successful.

In addition, naturally, the local focus of change management appears to become more critical. Van der Voet, Kuipers, & Groeneveld (2016) highlight that, despite the importance of organizational change for public management practice, organizational change is generally not studied as an implementation problem in public management research. Accordingly, public management research concerning organizational change is often focused on changes at the sector or national level, suggesting an extension to other spatial or functional levels.

3. The analytical triangle of the crisis, inadequate innovation, and incomplete change management

This paper argues that the three dimensions of “crisis-innovation-change management” although seemingly treated by the relevant literature as conceptually segregated and independent of each other, they constitute the most critical dimensions of development theorization. Therefore, this synthetic view can be particularly useful for the perception of the phenomenon of underdevelopment in local scale within the emerging “new globalization” phase of the world economy.

The global crisis is the current stage of the socio-economic “game” that spreads globally and in multiple operational dimensions nowadays. The global crisis calls into question its own balanced, unobstructed reproduction. Moreover, this structural and systemic process is not a superficial and sporadic state of affairs (Vlados, Deniozos, Chatzinikolaou, & Demertzis, 2018). In this context, an increasing number of “players” appear unable to fulfill their goals and meet their expectations in terms of their survival and development, and thus actively questioning the effectiveness of their broader strategy. At the same time, their former “appropriate behaviors” prove inadequate and ineffective in their specific field of action, while attempts to interpret the present and predict their future lead to widening failures.

The global crisis and restructuring is the era where old problems seem to come back to the forefront on every part of the globe; while new challenges emerge and spread rapidly (Bremmer, 2014; Lane, 2017). The current global crisis and restructuring thus appear to be at the same time both a period of the drastic overthrow of “yesterday's certainties” and a phase of redeployment of the entire global system in all its partial functional dimensions. In this approach, this sought-after “new” (in every space and partial implementation) that will provide a sustainable exit from the crisis is the birth of innovation (Carayannis, 2013). In this context, innovation is the implementation of new ways of resolving old or new problems in more efficient, effective, and “profitable” ways in a broader sense.

This study argues that the absence of innovation effectively implemented nourishes and expands the crisis: In this context, innovation is the only adequate way for the structural and long term exit from the crisis (OECD, 2009; Peris-Ortiz, Fuster-Estruch & Devece-Carañana, 2014). However, this “new” of every form and scope is the necessary dynamic condition that leads the “old” to the irreversible “death” by relieving the society as a whole of the pains of its “deathly agony” of the phase of the crisis. Moreover, this “new” in the course of world history seems always to delay its establishment. As a result, now is an era when the fruitless attempts to overcome the past repeat themselves for long periods. In this context, the expression of Antonio Gramsci (1971, p. 270) that “*The crisis consists precisely in the fact*

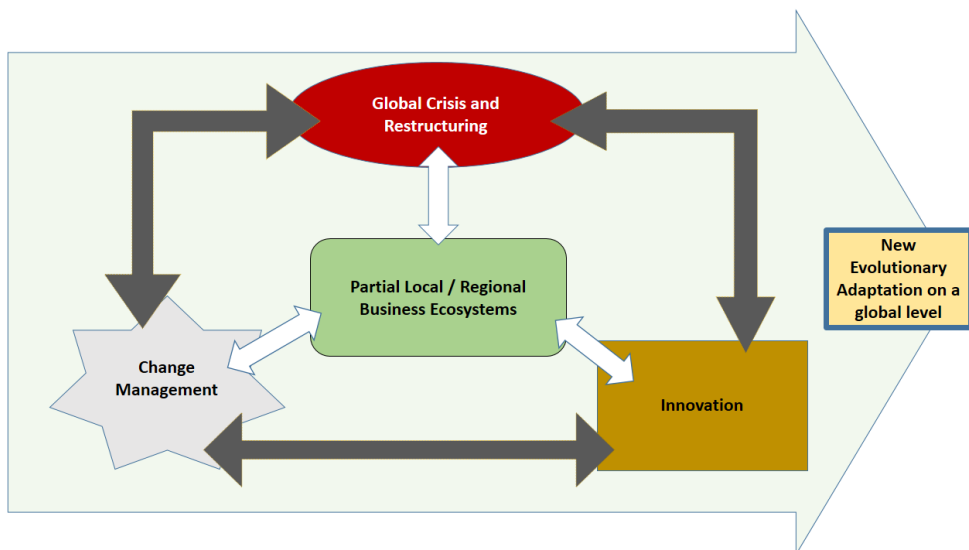
that the old is dying and the new cannot be born; in this interregnum, a great variety of morbid symptoms appear” seems particularly valid even today.

However, how does this “new” emerge, after all? Is it a birth coming after a “lucky coincidence” or is it the product of a more in-depth structural process and change?

In this analytical perspective, innovation (the “new”) seems to be simply the “fruit of the tree” of change management (Burnes, 2009; Worley & Mohrman, 2014). Namely, the chosen method each time to perceive, draw, combine, direct, coordinate and control, at each level of organized life, resources and strengths, in order to cope with evolving demands of the environment.

In this context, change management is the organic set of thoughts and actions, the mechanisms and methodologies developed (more or less organized and systematically) in order to adapt to the endless flow of change. In addition, when this flow of change creates drastic twists (i.e., when it appears as an outbreak of a crisis), the ability to manage change becomes the necessary and indispensable basis that opens the way (to a lesser or greater extent) for innovation. Thus, without effective change management, innovation always slows down while the efforts to overcome the “old” are unsuccessful and the specific strategic failures, technological, and managerial deficiencies perpetuate and reproduce the symptoms of the crisis (Vlados, Katimertzopoulos, & Blatsos, 2019).

FIGURE 3. THE NEW FORM OF EVOLUTIONARY ADAPTATION ON A GLOBAL LEVEL



Through this methodological perspective, managing change is not only connected with the sphere of innovation, but it is always closely tied up with the same dynamics of the crisis (Vlados, Deniozos, & Chatzinikolaou, 2018a). The crisis defines, therefore, the specific requirements of its overcoming, setting the specific limits of possibilities and the bar of

success (Birt, 2016; García, Maldonado, Alvarado, & Rivera, 2014) in all the mechanisms of change management, for all “actors” and for all eras (Figure 3).

Hence, the underlying assumptions and explanatory motor of the present study, in its key points, can be briefly described as follows:

- The three indivisible spheres of reality (global crisis, innovation, and change management) are always in historical coexistence and concurrence.
- There is always a particular historical-evolutionary character of the dynamics at the global socio-economic space (Boschma, 2004; Faissal Bassis & Armellini, 2018; Mack & Mayer, 2016) implemented simultaneously at all the structurally interconnected spatial levels: local, regional, national and supranational.
- The dynamics of overall socio-economic development define a complex, competitive architecture according to the partial local/regional ecosystems that try to adapt to this newly emerging reality evolutionarily (Bhawsar & Chattopadhyay, 2015).
- The necessary explanatory substratum of any partial development perspective at local and regional level appears now as a complex view of globalization studies. This perspective studies the innovative dynamics and the generated management of change by placing at the center the dynamics of local/regional socio-economic entrepreneurial ecosystems.
- entrepreneurial ecosystems.

4. The case of small and micro enterprises in Eastern Macedonia and Thrace via field research

4.1. The profile of the less-developed regional business ecosystem of Eastern Macedonia and Thrace

The region of Eastern Macedonia and Thrace is one of the thirteen regions of Greece. This region is a less-developed region in Greece and Europe (Boden, 2017). It brings together features of a border area, as all the regional units except Kavala are frontier. Border regions (Prokkola, 2019; Varol & Soylemez, 2019) combine socioeconomic, demographic and other problems, which contribute to these geographical units to fail more often to keep up with developments in other regions that occupy a more central geographical location. Thus, a border region may be (a) marginal due to reduced relations with other regions and the limited role played by economic, technological, cultural and political developments, and (b) disadvantageous because of the existence of inherent weaknesses which deprive developmental dynamism.

The fact that neighboring countries have a different social, economic, institutional, and political system is another cause of their regional nature. There is, therefore, a tendency to reduce the economic and administrative relations that develop naturally between neighboring regions, which often makes it challenging to develop cross-border cooperation (Medeiros, 2015; Miörner, Zukauskaitė, Trippl, & Moodysson, 2018).

TABLE 1. EASTERN MACEDONIA AND THRACE - REGIONAL FIGURES

GDP PER CAPITA. BY REGION, 2008 & 2016* (EUR, CURRENT PRICES)			
	2008	2016*	CHANGE %
Greece	21845	16378	-25.02%
Attica	29215	22204	-23.99%
North Aegean	17647	12266	-30.49%
South Aegean	24747	17769	-28.19%
Crete	19176	13811	-27.97%
Eastern Macedonia & Thrace	15568	11432	-26.56%
Central Macedonia	17437	12860	-26.24%
Western Macedonia	16748	14361	-14.25%
Epirus	14960	11785	-21.22%
Thessaly	16363	12662	-22.61%
Ionian Islands	21759	15182	-30.22%
Western Greece	16393	12058	-26.44%
Central Greece	19552	14727	-24.67%
Peloponnese	17224	13579	-21.16%

GROSS VALUE ADDED. BY INDUSTRY, 2008 & 2016* (EUR, CURRENT PRICES, IN MILLIONS).

Selected Industries: AFF: Agriculture-Forestry-Fishing, IC: Information-Communication, M: Manufacturing, C: Construction, TTAFS: Trade-Transportation-Accommodation-Food Services, FIA: Financial and Insurance Activities, PSTA: Professional, Scientific and Technical Activities

2008								
	AFF	IC	M	C	TTAFS	FIA	PSTA	SUM
Eastern Macedonia & Thrace	499	149	947	439	2094	205	323	4656
2016*								
Eastern Macedonia & Thrace	478	101	749	150	1205	173	215	3071

Source: Hellenic Statistical Authority. * Temporary data.

The observation that these regions are facing developmental problems is due to the following particularities:

- Isolation from leading centers of economic activity and decision-making
- Separation of their commercial centers from the natural hinterland, resulting in distortions in the form of trade and the provision of services
- Existence of limited natural resources (even though, in the case of Eastern Macedonia and Thrace this characteristic is not entirely applicable)
- Low level of social services to enterprises due to lack of staffing and limited technical infrastructure
- Problems of illegal immigration and smuggling
- Small economies do not allow the development of economies of scale

Low quality of transport infrastructure and other infrastructure elements.

The border regions usually lie at the end of the national transport chain and communication systems, components that contribute to the development deficiency of these areas (De Noni, Orsi, & Belussi, 2018). Not only that but the combination of elements of peripherality and isolation (Arieli, 2019; Mayer, Zbaraszewski, Pieńkowski, Gach, & Gernert, 2019) contributes to increased costs in a border area:

- The increased cost of money expressed as operating costs for the enterprises and the cost of living for residents,
- The increased cost of infrastructure and services due to negative economies of scale,
- Increased opportunity costs because there are fewer opportunities in the economic and other sectors,
- The increased cost of information because information flow arrives relatively late in remote areas and requires additional higher costs, particularly for obtaining specific information.

The Table 1 presents some quantitative data that capture the structural profile of the region of Eastern Macedonia and Thrace by taking as the base year the start of the rapid recession of the Greek economy in 2008 (Vlados, Deniozos, & Chatzinikolaou, 2018b).

The data show that the region of Eastern Macedonia and Thrace faced a substantial decrease of the GDP per capita between 2008 and 2016 (-26.56%), a more considerable decrease compared with the rest of country (-25.02%). For 2016, the region recorded the lowest per capita GDP of the country (just €11,432). At the same time, a change by 34% in gross value added in selected productive sectors occurs between 2008 and 2016 in the whole region (from €4,656 million to € 3,071 million). The data illustrate that the productive sector “Trade-Transportation-Accommodation-Food Services” has the largest share in the gross value added of the region.

Moreover, the Greek Regional Aid Map for the period 2014-2020 adopted by the European Commission on May 7th, 2014 [C (2014)2642-State aid No. SA.38450 (2014/N), valid until 31/12/2020], classifies this border region “A” because its GDP per capita is less than 75% of the EU average [Article 107(3) of the Treaty on the Functioning of the EU] and is eligible for regional investment aid. Accordingly, the amended Regional Aid Map, the rates of which are 35% when the size of the enterprise is large, 45% for medium-sized enterprises and 55% for small-sized enterprises, apply to all investment projects approved from 01/01/2017 to 31/12/2020 (Delhetheus, 2018, pp. 382-383).

4.2. The case of small and micro enterprises in the region of Eastern Macedonia and Thrace: field research and findings

To understand the analytical triangle of “crisis-innovation-change management” qualitatively, the authors conducted a field survey in a sample of small and medium-sized enterprises within the less-developed regional socio-economic system of Eastern Macedonia and Thrace.

4.2.1. The identity of the field research

This survey interviewed a non-weighted sample of 48 micro and small private enterprises operating in the region of Eastern Macedonia and Thrace and obtained data during the winter of 2017. One of the selection criteria was for these companies to employ a workforce of up to 50 employees. The field research did not take into account the particular sector of economic activity, although the majority of the enterprises happened to operate in the food and beverages sector of economic activity.

This field research aimed to investigate how the owner or another member of a small enterprise perceive the dimensions of the crisis, innovation, and change management. This field research is exploratory and qualitative research (Shields & Rangarajan, 2013), not arrived at employing statistical procedures or other means of quantification (Strauss & Corbin, 1990). In particular, this qualitative research (Baxter & Jack, 2008; Neergaard & Ulhøi, 2007) has only an interpretive character, aimed at discovering the meaning events have for the individuals who experience them, and the interpretations of those meanings by the researcher (Hoepfl, 1997). According to Rowley (2012, p. 262), who defines a semi-structured interview as *“an interview schedule that centers on around six to 12 well-chosen and well-phrased questions to be delivered mostly in a set order,”* the interviewer asked general and semi-structured questions in a sample of enterprises. The aim was to study, through personal interviews that reflect the views and assessments of the respondents, how the members of small and micro enterprises perceive the phenomena of crisis, innovation, and change management.

The responsible interviewer initially asked the member of the firm to record the name, the subject, the number of employees, the address, and contact details. Then, the interviewer had to take and record an interview with the member of the enterprise, who had to answer the following questions¹:

About the **perception of crisis**:

- i. “How does your enterprise perceive this crisis?”
- ii. “What are the effects of the crisis on your enterprise?”
- iii. “What do you believe are the root causes of the current crisis?”
- iv. “What do you think are the solutions for overcoming the crisis?”

About the **perception of innovation**:

- v. “How does your enterprise understand the concept of innovation?”
- vi. “In what way and how do you think can innovation help your enterprise?”
- vii. “In what specific ways has your enterprise innovated recently?”
- viii. “What do you think are the main barriers to innovation in your enterprise?”

¹ The average response time of all questions ranged in approximately 15 minutes per interviewee, and the raw recorded and transcribed material is available to any interested researcher upon request.

About the **perception of change management**:

- ix. “How does your enterprise understand the concept of change (What does it feel to change, above all)?”
- x. “How does your enterprise understand the concept of adapting to change?”
- xi. “How does your enterprise understand the concept of managing change?”
- xii. “In what ways does your enterprise manage change?”

4.2.2. Analysis of findings

Initially, based on the answers to the four crisis questions and regarding the respondents' estimates, the following conclusions are extracted:

- I. Concerning how the enterprise perceives the current crisis, the vast majority of responses highlight the decline in the company's turnover and profits and the overall fall in market prices. A typical answer, which primarily reflects the majority, is that *“our revenues are less than earlier years.”*

The majority of responses perceive the economic aspect of the crisis merely as a reduction in the customer's purchasing power by also emphasizing the overall social problem where *“a multitude of people cannot provide themselves with the essentials for living”* or expressing pessimism with expressions like *“we have not reached the bottom yet.”*

On the other hand, a significant minority stresses that the crisis has not influenced the business, arguing that *“if you are good at what you are doing, there is no crisis”* or *“the crisis has not undermined us at all because we are doing a good job.”*

Also, a notable minority gave responses such as *“I think that this crisis debate is more on paper than in reality”* or *“we are enemies for the state”* or *“in order to make up for the lost turnover for the necessary survival of our business we need to think about other ways and extend to other business activities.”* This kind of responses indicates that there is a relative ambiguity about the very meaning and importance of the current crisis.

- II. Regarding the effects of the crisis on the enterprise, the vast majority of responses, in the same manner to the first question, highlights the reduced turnover and leaves the question of their reduced profitability to a much lower position. Many of the respondents report the effects of the crisis upon the business. They point to dimensions such as the reduction of staff, the longer work hours for the owners, the high taxation, the hostile attitude of the state, the lack of liquidity, the lack of a variety of suppliers, and finally the limited access to financing programs and bank lending.

It is worth noting that the majority of responses to the first and second question are overlapping, giving similar responses: i.e., the majority of respondents do not seem capable to distinguish between the perception and the results of the crisis upon the business.

Also, a notable minority gave responses such as *“if you want to keep your quality standards, then you have to make cuts in staff and other aspects of the business.”* Another stated that *“the management of our company took some quick decisions in 2008 when it saw the crisis ... the company decided to reduce its investment program without of course reducing the quality and quantity of production.”* Other responses seemed to deny the reality arguing that *“I am not facing any crisis, and I am satisfied with the results of the business.”*

III. About what they believe are the root causes of the current crisis, the vast majority of respondents blame the governmental agencies and policies. In particular, they highlight the political corruption and poor management in public finances as the critical source of the crisis. To a lesser extent, the respondents highlight the excessive borrowing from the part of citizens during recent years.

However, a significant minority of respondents seem unable to distinguish between the causes and effects of the crisis, as they perceive as causes of the reduced income and high taxation the closure of many small businesses and the high unemployment.

At the same time, a notable minority gave responses such as *“we have not realized that productivity defines how profitable you are”* or *“everyone says it is a debt crisis, but ultimately it is a crisis of attitude, a moral crisis and a cultural crisis”* or *“the cause is the lack of productive capacity of the country.”*

IV. Regarding what they believe are the critical solutions to overcome the crisis, the vast majority of responses stressed the need for government interventions, focusing on tax cuts, debt reduction, the need to increase public investment, and, more generally, government expenditure. A typical answer in this direction is that *“We cannot do much. Politicians have to find the right solutions.”*

A significant minority, however, identifies as a solution to better organizing the “modus operandi” in order to improve competitiveness and make greater use of the private initiative. In this direction, there are typical responses such as *“a basic solution for overcoming the crisis comes from the rightful and healthy business”* or *“only if we look ahead and improve our production we can have a good result.”*

Interestingly, a notable minority gave responses such as *“our country has become a subordinate of external institutions and geopolitical interests that do not want to get out of the crisis”* or *“we have to exploit our country’s wealth in oil reserves”* or *“the public sector should make deep cuts to stop corruption.”*

Secondly, based on the answers to the four innovation questions and regarding the respondents’ estimates, the following conclusions are reached:

V. On how does the enterprise perceive the concept of innovation, the vast majority of responses perceive innovation as necessary, giving definitions related to the availability of technological applications (Internet, social media) which would improve the current performance of the business (mainly increase in turnover).

It seems that the majority of these locally-established small enterprises perceive the correlation of innovation with generating profits, but limited to the concept of innovation as something “new” almost exclusively on the product level. In this view, the “discovery” of the

new arises spontaneously from practical experience and friction with customers, in the absence of a systematic framework for understanding and promoting innovation at all operational levels of the organization.

A significant minority, however, notes that innovation is something quite distant and elusive, the demand for which in the present “circumstances” of the crisis will probably only intensify competition and lead to loss of profits.

Interestingly, some sporadic responses point out that innovation can be *“opening up to new management strategies”* or *“a change at all fronts: a movement directed forward”* or that *“innovation is the offer of a product/service that you cannot easily find on a wide market scale.”*

VI. Regarding on how they think innovation could help their enterprises, the vast majority of entrepreneurs refer to the “facilitation” that specific technological applications (energy, online, mechanical) provide for limiting the personal work of the business owner.

It is worth emphasizing that the majority of responses to this question showed unable to perceive any difference between the first and second question since the answers are overlapping significantly. They fail to distinguish between the definition of innovation and specific systematic ways that can lead to it.

A significant minority, however, seems to realize that innovation is an improvement in the business organization, focusing mainly on the aspect of company sales and end-user satisfaction, showing a willingness to increase their clientele.

Also, a notable minority gave responses such as *“innovation helps you stand out”* or *“assists every business to be modern”* or *“the adoption of innovation at every stage in the business operation as well as in products and processes is helpful for the development.”*

VII. About the specific ways in which their enterprise has innovated recently, the vast majority of respondents appear divided into two categories: the businesses referring to the introduction of new end-products as a critical innovation activity and the businesses tending to face the current crisis as a severely disruptive process for business development that cancels any innovative effort.

A significant minority also notes that their enterprise has innovated recently by introducing technological applications or machinery and, more generally, by upgrading and refurbishing its customer reception area (store).

Interestingly, some sporadic responses point out that *“innovation is a new product”* or that *“we have put emphasis recently on production and to create quality products to be competitive in the global environment we live in.”* Alternatively, *“we just sell coffee, we have not discovered anything special”* or that *“we have innovated by introducing product codes different than our competitors’ at least in our region”* or that *“at this point, there is stagnation and I cannot talk about many innovative things. Everything I did was before 2010”*.

VIII. In relation to the main obstacles to innovation in their enterprise, the vast majority of the respondents focused on the lack of financial resources (typically, they noted factors such as financial difficulties, the public sector with the over-taxation of entrepreneurship, the economic crisis, the legal framework, lack of liquidity, lack of capital and borrowing).

A significant minority, however, point out that it is not so much the external factors that affect the operation of the business and its innovation, as much as the entrepreneur's approach and attitude.

Some minority responses in this direction appear more "progressive" by stating that *"the main obstacles come from the obsession with old selling techniques, old merchandise management, that is, everything that is attached to the old one."* Alternatively, *"normally there should be no obstacles for a professional, you have to set small goals and slowly accomplish them"* or *"the obstacle is knowledge. If there is an obstacle to innovation is to find the man who will bring knowledge."*

Based on the answers to the four change management questions and regarding the respondents' estimates, the following extracted conclusions are useful:

IX. In connection with how the company understands the concept of change (what feels that mainly changes), the vast majority of responses focused on the changes brought by the crisis in the organization: the reduction in turnover and profit, increasing pressure for better quality and, at the same time, a continually demanding price squeeze.

However, a significant minority sees the change positively, arguing that change provokes good business and which should not be static. A typical response in this direction is that *"I do not feel that anything has changed in business for the worst, just because of the crisis."* At the same time, some minority answers seem not even understand the question and the concept of change, not managing to respond in an integrated manner.

X. As to how the enterprise perceives the concept of adaptation to change, the vast majority seems to try adapting to the changes brought by the crisis by cutting expenses (mainly by reducing staff and wage cuts) to achieve lower prices for their products and services. They, therefore, show that they perceive the change passively: emphasizing that *"we did not choose to change, either way, we have to adjust."*

A significant minority, in terms of the number of received responses, also emphasizes psychological factors, saying that *"you need to adapt passionately, manage your feelings, and try continually and with dedication."* Some other minority responses emphasize that *"we are adapting by pushing our prices down and, at the same time, to succeed in achieving innovation,"* or *"we need to be flexible, to adapt to situations in order for the business to be profitable so we can face the crisis."*

XI. Regarding how the enterprise understands the concept of change management, the vast majority of responses focus on managing change as a cut in the company's staff, and a squeeze of its operating costs. Typically, the answers highlight in several cases that *"by reducing staff we can survive and we are now waiting for the positive*

elements to respond to new market demands.” At the same time, the majority of replies note that managing change is challenging, mainly due to the limited profitability of the company.

A significant minority appears to approach change management more thoroughly, focusing on its organizational dimension. Specifically, the typical responses are: *“to change the organizational chart a bit”* or *“to implement an excellent plan so that we can manage all that change”* or *“with flexibility and adaptability some people in each field have to undertake initiatives and leadership.”* Some interesting minority responses also note that *“we are years in this pattern, we are just adjusting according to our suppliers”* or *“managing change means I do not leave to luck what I want to succeed.”*

XII. Concerning the ways that the enterprises manage the change, the vast majority of responses focus on cost containment issues, with virtually no reference to using systemic methods to increase operational efficiency. Typically, they stress that *“we are waiting for the state's support”* or that *“fewer people are doing fewer jobs”* or *“cutting down the costs as much as we can.”*

A significant minority, however, refers to methods such as *“continuous education and interactive seminars”* or *“we need to develop a strategy”* or *“to keep track of developments in a scientific context.”* At the same time, some responses stress that their company has improved *“by offering things that would not need to offer them otherwise to be able to deal with the change that has created the crisis.”* Also, some minority responses indicate that *“we have made no significant changes to our business”* or *“the company manages change with caution, attention, study, and careful steps.”*

5. Conclusions, limitations, and prospects of expanding the research

The purpose of this research was to highlight how a sample of small and micro enterprises hosted in a less-developed regional business ecosystem (the Greek region of Eastern Macedonia and Thrace) perceive the spheres of the crisis, innovation, and change management.

From the field research that the authors carried out in this region, the following conclusions synthesize upon the knowledge acquired from the scientific literature and the answers made by the respondents concerning the “crisis-innovation-change management” analytical triangle:

- A. The way these businesspeople perceive the phenomenon of crisis in the field is far from the fundamental approaches and findings of the existing literature. The majority of the responses highlighted a shallow perception of the crisis as a strictly “exogenously determined” process (Amable, 2017). Accordingly, the businesspeople of the sample perceived the crisis as dominated by financial terms and, above all, as something that compresses their turnover and the immediate purchasing power of their customers. The majority of these businesspeople did not perceive the structural dimensions related to the crisis and especially the low competitiveness of this regional socio-economic subsystem.

- B. The way these businesspeople perceive innovation appears to have a predominantly sporadic and reflective character. This concept of innovation focuses, almost exclusively, on superficial condition of “technical improvement” without any more profound insistence on the parallel organizational and socio-economic dimensions and requirements (Anderson, Potočník, & Zhou, 2014; Damanpour & Aravind, 2012). The majority of the small enterprises that took part in this present research seem to perceive innovation as an external factor that they cannot either control or determine. As such, they are unable to distinguish the global socio-economic reasons and processes that make their particular regional business ecosystem weak in terms of production, diffusion, assimilation, and expansion of their innovative potential.
- C. The way these businesspeople perceive change management in the studied field also seems superficial and shortsighted. Their perception of change management is merely reflective because the “conjuncture” of the crisis forces their companies into “tactical folds” and retreats and, more generally, “passively” responses (Bennett, 2002; Laubacher & Malone, 2003). Thus, in the vast majority of cases, the applied change management method involves only measures of compressing product costs and limiting operating costs, without any greater ambition to improve business efficiency over time systematically.
- D. Based on the previous observations, the entrepreneurs of these micro and small enterprises in this field research demonstrated difficulty in understanding the evolutionary interconnections between the crisis, inadequate innovation, and incomplete management of change that to the reproduction of underdevelopment in their business ecosystem.

In conclusion, this study has helped to validate the gap between the scientific view of the crisis, innovation, and change management and the perception of businesspeople daily practice in small and micro enterprises inside a less-developed regional business ecosystem. At the same time, it offers a first empirical qualitative mapping to refine this field research in the future.

In conclusion, the authors must point out that this is a limited analysis of interpretation resources and prediction because of the non-weighted sample and the “introductory character” of the field research. In a weighted sample of enterprises, either in terms of size or in terms of industry, or a combination of the two, a field survey could draw more operational conclusions about how the businesspeople of everyday practice perceive the “crisis-innovation-change management” analytical triangle within a less-developed regional business ecosystem. The authors could reassess their hypothesis in the field through structured questionnaires that the evolutionary adaptation of a less-developed regional socioeconomic business ecosystem is a matter of the co-evolving conditions of the crisis, innovation, and change management.

Therefore, to cover specific research gaps in the future, this research could explore with greater completeness where the less-developed regional business ecosystems are lagging in terms of perceiving the “crisis-innovation-change management” analytical triangle and how this perception concerns the generated phenomenon of regional underdevelopment.

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